Sairohith Vamsani

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Summary

Final-year B.Tech Computer Science (AI & ML) student with hands-on experience in machine learning, NLP and data analysis using Python and R. Seeking AI/ML, data science or software roles to build scalable, data-driven solutions.

Education

Malla Reddy University — Hyderabad, India
B.Tech, Computer Science and Engineering (AI & ML) — CGPA: 8.64/10 — *2022 – 2026*

Experience

Infosys Springboard Virtual Internship 6.0 — Sep 2025 – Nov 2025
Artificial Intelligence and Machine Learning Intern — *Virtual*

- Contributed to the **Semantic Search and Query Module** of a healthcare analytics platform
- Applied **NLP** techniques to improve query relevance and search accuracy
- Worked with **Python** and **knowledge graph**-based retrieval systems
- Supported testing and optimization of scalable project modules

Projects

HealthIntellect – Medical NLP Analytics Suite | *Python, spaCy, BioBERT, Neo4j, Streamlit*

- Built a Streamlit platform that turns medical text from **Wikipedia and arXiv APIs**, typed text and uploaded files into a **medical knowledge graph**
- Implemented **hybrid NER** combining spaCy, a **BioBERT** pipeline and regex patterns, with entity deduplication
- Extracted relations by dependency parsing, semantic and medical-pattern methods, with confidence scoring, and formed knowledge triples
- Built a **semantic search** module with Sentence-Transformers (all-MiniLM-L6-v2) and cosine similarity over documents and triples
- Exported graphs to **Neo4j**, visualised them with NetworkX and Plotly, and added SQLite user accounts with **JWT** login, roles and an admin dashboard

Disease Insight – Multiple Disease Prediction System | *Python, XGBoost, Streamlit, Pandas, NumPy, Joblib*

- Developed a web-based healthcare platform that predicts **multiple diseases** from real-time user inputs using **XGBoost**
- Performed data preprocessing, feature scaling and feature selection to improve model accuracy
- Built an interactive **Streamlit** interface for real-time predictions and clear result display
- Deployed models with **Joblib** for fast loading and inference, in a modular design that supports adding new disease modules

Technical Skills

Languages: Python, R, SQL
AI / ML: Machine Learning, Natural Language Processing (NLP), XGBoost, Prompt Engineering, LLM Evaluation
Libraries: Pandas, NumPy, spaCy, Hugging Face Transformers, Sentence-Transformers, Streamlit, Plotly, Joblib
Databases: MySQL, Neo4j, SQLite
Tools: Power BI, Microsoft Excel, VS Code, Jupyter Notebook
Data: Data Collection, Data Cleaning, Data Visualization, Statistics

Certifications

Fundamentals of Machine Learning | *Microsoft Learn* — Feb 2025
Fundamentals of Generative AI | *Microsoft Learn* — Jan 2025
Introduction to Python Programming | *University of Pennsylvania, Coursera* — Jun 2023